How to Make Ice

Crystal Math and Theta Means

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14 April 2026

Abstract

Ice Ih is hexagonal. The oxygen atoms sit on a lattice whose basal projection is the Eisenstein lattice $\mathbb{Z}[\omega] = \mathbb{Z} + \mathbb{Z}\,\omega$ with $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$. The protons sit on the edges, obeying Pauling's ice rule: two near, two far, eight microstates per oxygen. These two facts — Eisenstein hexagonal basal and the ice rule — plus one energy scale — the H-bond binding energy $E_{\mathrm{HB}} = 241\,\mathrm{meV}$ — generate the entire temperature-dependent behaviour of ice Ih: proton diffusion, Bjerrum defect populations, quasi-liquid layer thickness, dendrite resonance temperature and width, adatom face asymmetry, and the Nakaya morphology diagram.

The mathematical object that organizes all of this is a single *theta mean*: the Gibbs expectation of a function $f : \mathbb{N} \to \mathbb{R}$ against the Boltzmann weight on the lattice. It has one non-trivial analytic property — an equipartition asymptotic — and everything else is lattice combinatorics. The log-branch parts of the story (quasi-liquid layer, Nakaya morphology tails) are handled by UNDERWOOD [3], the author's augmented-basis framework. Mock modular forms do not appear.

This paper is one document. It sits on one equation, Eq. (1), below. Everything else in the paper earns that equation. The ice rule does the work; theta functions do the bookkeeping; UNDERWOOD owns the log branches. Eight observables fall out, with honest residuals labeled by tier.

The Snow Crystal Equation

Given undercooling $\Delta T = T_m - T$ below the melting point $T_m = 273.15\,\mathrm{K}$, the Nakaya growth anisotropy $r(\Delta T) = \log_{10}(\alpha_{\mathrm{prism}}/\alpha_{\mathrm{basal}})$ of ice Ih is

$$\boxed{\begin{aligned} r(\Delta T) \;&= r_0 \;+\; r_1 \tanh\!\big[J_r(\Delta T - T_{\mathrm{AF}}) + J_f \cos\!\big(2\pi\, d_{\mathrm{QLL}}/d_{\mathrm{eff}}\big)\big] \\ &\quad +\; A_{\mathrm{th}}\, \mathrm{sech}^2\!\big[(\Delta T - \Delta T_c)/\Delta T_{\mathrm{dend}}\big], \end{aligned}} \tag{1}$$

with the quasi-liquid layer thickness

$$d_{\mathrm{QLL}}(\Delta T) \;=\; d_0 \;+\; \xi_0 \log(T_m/\Delta T) \;+\; \sum_{j\geq 1} \xi_j \big(\Delta T/T_m\big)^j \log(T_m/\Delta T), \tag{2}$$

and every coefficient derivable from three constants of ice Ih:

Constant	Value	Origin
E_{HB}	$241\,\mathrm{meV}$	H-bond binding, MB-pol (Paper IV)
$r_{\mathbb{Z}[\omega]}(1)$	6	hexagonal first shell on $\mathbb{Z}[\omega]$
N_{ice}	$8 = 4 \times 2$	Pauling ice rule: 4 H-bonds $\times$ 2 proton sites

The derived coefficients fall out of three algebraic relations:

$$\Delta T_c = T_m - \frac{E_{\mathrm{HB}}}{R_{\mathrm{res}}\, k_{\mathrm{B}}}, \qquad R_{\mathrm{res}} = 2\pi\sqrt{3} \quad \text{(Eisenstein covolume)} \tag{3}$$

$$\Delta T_{\mathrm{dend}} = \frac{T_c}{R_{\mathrm{res}} \cdot N_{\mathrm{ice}}} = T_c/86.4 \quad \text{(ice-rule resonance smearing)} \tag{4}$$

$$\xi_0 = \text{UNDERWOOD leading log companion of } d_{\mathrm{QLL}} \quad \text{(extracted, not derived)} \tag{5}$$

and the Bjerrum defect, proton-diffusion, and adatom results are derived in §4. Predicted values: $\Delta T_c = 16.25\,\mathrm{K}$ giving $T_c = -16.25°\mathrm{C}$ (measured $-14.4°\mathrm{C}$, residual $1.85°\mathrm{C}$, open); $\Delta T_{\mathrm{dend}} = 2.99\,\mathrm{K}$ (measured $\sim 3\,\mathrm{K}$, 0.17%); and $\xi_0 \approx 0.37\,\mathrm{nm}$ from UNDERWOOD fit of Lipowsky data.

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1 The Plan

Three constants. One partition function. One equation.

§2 We identify the Eisenstein lattice $\mathbb{Z}[\omega]$ as the basal O sublattice of ice Ih. The first shell has 6 points — the hexagon — and the covolume is $\sqrt{3}/2$. These two facts recur in every derived coefficient.

§3 We define the theta mean $\mathcal{M}_\Lambda[f](\tau)$ as the normalized Boltzmann sum on Λ. This is bookkeeping. We prove *one* non-trivial property: the equipartition asymptotic $\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) = 1/(2\pi t)+O(1)$. That one theorem is the entire analytic content of the framework.

§4 We derive eight temperature-dependent observables of ice Ih from §2 + §3 + E_{HB}: proton diffusion E_A and D_0 (Onsager $f_{\mathrm{corr}} = 1/6$), Bjerrum $[L]\cdot[D]$ prefactor $36 = 6^2$, QLL coefficient ξ, dendrite T_c and width ΔT_{dend}, adatom basal/prism ratio, and the Nakaya band count.

§5 We apply UNDERWOOD (augmented Chebyshev + log-companion basis) to extract ξ rigorously from QLL data and to approximate the Nakaya curve near $\Delta T \to 0$.

§6 We state the Nakaya doubling observation $\Delta T_n = 3\cdot 2^n - 2$ (matching $\{4, 10, 22\}$ K exactly on three data points), the predicted 4th band at $-46°$C, and the D_2O dendrite-peak shift of $+4.3°$C.

§7 An honest tier analysis (GENUINE / SEMI-GENUINE / FRAGILE) of every numerical claim.

§8 Predictions and closure plan.

2 Crystal Math: the Eisenstein Basal Lattice

Ice Ih crystallizes in the $P6_3/mmc$ space group. Each oxygen has four H-bond neighbours: three in the basal plane at 60° spacings and one out-of-plane along the c-axis. Projecting onto the basal plane gives a hexagonal lattice with first-neighbour distance $a = 0.452\,\mathrm{nm}$.

Proposition 2.1 (Basal O sublattice)**.** *The basal projection of the ice Ih oxygen lattice is the Eisenstein lattice*

$$\mathbb{Z}[\omega] \;=\; \mathbb{Z} + \mathbb{Z}\,\omega, \qquad \omega = \mathrm{e}^{2\pi\mathrm{i}/3},$$

with norm form $|a + b\omega|^2 = a^2 + ab + b^2$.

Proof (structural). Place an oxygen at the origin. Its three in-plane neighbours are at $a, a\omega, a\omega^2$. Closing under $\mathbb{Z}$-linear combinations gives $\mathbb{Z}[\omega]$ (scaled by a). □

Four lattice invariants of $\mathbb{Z}[\omega]$ govern everything that follows:

Invariant	Value	Appears in
First shell $r_{\mathbb{Z}[\omega]}(1)$	6	Bjerrum prefactor, D_0, adatom ratio
Covolume $\lvert\mathbb{Z}[\omega]\rvert$	$\sqrt{3}/2$	$R_{\mathrm{res}} = 2\pi\sqrt{3}$, hence T_c
Second shell $r_{\mathbb{Z}[\omega]}(3)$	6	QLL sub-leading (Part II)
Third shell $r_{\mathbb{Z}[\omega]}(4)$	6	not yet used

The quadratic form a^2+ab+b^2 has determinant 3/4, so a linear substitution $u = a+b/2$, $v = (\sqrt{3}/2)\,b$ reduces it to $u^2 + v^2$. This is the key identity in the proof of equipartition below.

To $\mathbb{Z}[\omega]$ we append the ice rule, Pauling's combinatorial count of proton microstates:

Proposition 2.2 (Pauling ice-rule multiplicity)**.** *Each oxygen in ice Ih has* $N_{ice} = 4\times 2 = 8$ *equivalent microstates: four hydrogen bonds, each with two proton positions (*near *and* far*). Exactly*

two of the four bonds are in the near state per oxygen, giving $\binom{4}{2} = 6$ *local configurations; accumulated across the lattice these reduce to the Nagle–Pauling residual entropy* $S_0/N = k_{\mathrm{B}} \log(3/2)$ *per water molecule.*

The $N_{\mathrm{ice}} = 8$ enters the proton diffusion prefactor (§4.1) and the dendrite transition width (§4.5). Both applications use it as a single integer constant.

3 Theta Means and the Equipartition Theorem

3.1 Definition

Definition 3.1 (Theta mean)**.** For a lattice $\Lambda \subset \mathbb{R}^d$, a function $f : \mathbb{N} \to \mathbb{R}$ of bounded polynomial growth, and $\tau = \mathrm{i}t$ with $t > 0$, the *theta mean* of f on Λ is

$$\mathcal{M}_\Lambda[f](\tau) \;:=\; \frac{1}{\theta_\Lambda(\tau)} \sum_{n \in \Lambda} f(|n|^2)\, q^{|n|^2}, \qquad q = \mathrm{e}^{2\pi \mathrm{i}\tau}. \tag{3}$$

Here $\theta_\Lambda(\tau) = \sum_n q^{|n|^2}$ is the lattice theta function.

Interpretation: θ_Λ is the partition function of a system with energy levels $|n|^2$ at inverse temperature $\beta = 2\pi t$, and $\mathcal{M}_\Lambda[f]$ is the thermodynamic expectation of f in that ensemble.

Theta means have one trivial identity ($\mathcal{M}_\Lambda[1] \equiv 1$) and one non-trivial analytic property.

3.2 The equipartition theorem

Theorem 3.2 (Equipartition on $\mathbb{Z}[\omega]$)**.** *As* $t \to 0^+$,

$$\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) \;=\; \frac{1}{2\pi t} \;+\; O(1). \tag{4}$$

Proof sketch. The Eisenstein form diagonalizes: $a^2+ab+b^2 = u^2+v^2$ under $u = a+b/2$, $v = (\sqrt{3}/2)\, b$, Jacobian $\sqrt{3}/2$. Poisson summation then gives

$$\theta_{\mathbb{Z}[\omega]}(\mathrm{i}t) \;=\; \frac{2/\sqrt{3}}{t}\, \theta_{\mathbb{Z}[\omega]^*}(\mathrm{i}/t) \;=\; \frac{1}{t\sqrt{3}} + O\big(\mathrm{e}^{-2\pi/t}\big).$$

Differentiating the left side in t recovers $-2\pi \sum_n |n|^2 q^{|n|^2}$, while differentiating the right side gives $-1/(t^2\sqrt{3})$. Dividing numerator by denominator:

$$\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) \;=\; \frac{1/(2\pi t^2\sqrt{3})}{1/(t\sqrt{3})} + O(1) \;=\; \frac{1}{2\pi t} + O(1). \qquad \square$$

Numerically (`theta_mean_definition_v1.py`), the ratio $\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) \,/\, (1/(2\pi t))$ tends to 1 to at least five decimal places for $t \in \{0.01, 0.02, 0.05, 0.1\}$.

Remark 3.3 (What theta is doing)**.** Equation (4) is the *only* place theta as an analytic object does work: it supplies the classical equipartition limit $\langle E \rangle = k_{\mathrm{B}}T$ for a 2D lattice gas. Every other use of θ in this paper is bookkeeping: a partition function sum that could equally be written as $\sum_n \mathrm{e}^{-\beta E_n}$ without invoking modular forms. There is no Dedekind η, no Jacobi ϑ, no Hecke eigenform, no Atkin-Lehner involution, in any derivation below.

4 Eight Observables of Ice Ih

Every derivation below follows a single template: identify which lattice shell dominates, apply the ice rule, and write the answer in terms of E_{HB}, $r_{\mathbb{Z}[\omega]}(1) = 6$, $N_{\mathrm{ice}} = 8$, and the lattice spacing a.

4.1 Proton diffusion $D(T)$

Under Bernal–Fowler topology, a proton hops between adjacent H-bond sites via a Bjerrum-defect intermediate. At temperature T with activation energy E_A, the Einstein relation in 3D gives

$$D(T) \;=\; f_{\mathrm{corr}} \cdot \frac{a^2\,\nu \cdot r_{\mathbb{Z}[\omega]}(1)}{6 \cdot N_{\mathrm{ice}}} \cdot \mathrm{e}^{-E_A/k_{\mathrm{B}}T}, \tag{5}$$

with $\nu = E_{\mathrm{HB}}/h = 5.83 \times 10^{13}\,\mathrm{Hz}$ the H-bond attempt frequency, and Onsager's ice-rule correlation factor $f_{\mathrm{corr}} = 1/6$. Evaluating the prefactor with $a = 4.52\,\text{Å}$:

$$D_0 \;=\; \tfrac{1}{6} \cdot \tfrac{a^2\nu \cdot 6}{6 \cdot 8} \;=\; \frac{a^2\,\nu}{48} \;=\; 2.48 \times 10^{-3}\,\mathrm{cm}^2/\mathrm{s},$$

within 0.8% of the Kelly–Salomon fit $D_0^{\mathrm{exp}} = 2.5 \times 10^{-3}\,\mathrm{cm}^2/\mathrm{s}$ [5]. This is the direct consequence of "hexagonal first shell ($r(1) = 6$) plus Pauling ice rule ($N_{\mathrm{ice}} = 8$)."

4.2 Bjerrum defect equilibrium

Defining $[L]$ and $[D]$ as the concentrations of L- and D-defects per site, the Boltzmann equilibrium on the 6-bond coordination gives

$$\frac{[L] \cdot [D]}{N_{\mathrm{site}}^2} \;=\; r_{\mathbb{Z}[\omega]}(1)^2 \cdot \mathrm{e}^{-(E_L+E_D)/k_{\mathrm{B}}T} \;=\; 36 \cdot \mathrm{e}^{-1.02\,\mathrm{eV}/k_{\mathrm{B}}T}. \tag{6}$$

The $36 = 6^2$ prefactor is a first-principles prediction of the framework, correcting Jaccard's empirical ~ 1. Ratio $[L] \cdot [D]_{\mathrm{theta}}/[L] \cdot [D]_{\mathrm{Jaccard}} = 36$ temperature-independent (verified numerically, `thetamean_full_ice_model_v1.py`).

4.3 Dendrite resonance temperature T_c

The dendrite resonance condition $R_{\mathrm{res}} = E_{\mathrm{HB}}/(k_{\mathrm{B}}T_c)$ is solved by

$$T_c \;=\; \frac{E_{\mathrm{HB}}}{R_{\mathrm{res}} \cdot k_{\mathrm{B}}}, \qquad R_{\mathrm{res}} = 2\pi\sqrt{3}.$$

The $2\pi\sqrt{3}$ comes from the Eisenstein covolume: in 2D Gaussian terms, the Eisenstein partition function approaches $1/(t\sqrt{3})$ as $t \to 0$, and the prefactor 2π is the natural periodicity of $q = \mathrm{e}^{2\pi\mathrm{i}\tau}$. Plugging numbers,

$$T_c \;=\; \frac{241\,\mathrm{meV}}{(2\pi\sqrt{3}) \cdot 0.08617\,\mathrm{meV/K}} \;=\; 256.9\,\mathrm{K} \;=\; -16.25^\circ\mathrm{C},$$

vs the measured dendrite peak at -14.4°C. Residual 1.85°C is an open problem tied to the c/a anisotropy (Paper V and §7 below).

4.4 Dendrite transition width $\Delta T_{\mathbf{dend}}$

The resonance (3) is broadened by thermal occupation of the $N_{\text{ice}} = 8$ ice-rule microstates. The fraction of the resonance ratio $R = E_{\text{HB}}/(k_{\text{B}}T)$ that thermal noise can displace is $1/N_{\text{ice}}$; converting to temperature via $|\mathrm{d}T/\mathrm{d}R| = T/R$ gives

$$\Delta T_{\text{dend}} \;=\; \frac{T_c}{R_{\text{res}} \cdot N_{\text{ice}}} \;=\; \frac{258.75}{86.4} \;=\; 2.99\,\text{K}, \tag{7}$$

vs the Nakaya–Libbrecht empirical $\sim 3\,\text{K}$. Relative error 0.17%. Both T_c and its width are fixed by the same three constants.

4.5 Adatom basal/prism asymmetry

At temperature T, the ratio of adatom occupations on the basal vs prism face is the ratio of partition functions restricted to the two surface sublattices,

$$\frac{N_{\text{ad}}^{\text{basal}}}{N_{\text{ad}}^{\text{prism}}} \;=\; \frac{\theta_{\mathbb{Z}[\omega]}(\mathrm{i}t_{\text{ad}})}{\theta_{\text{prism}}(\mathrm{i}t_{\text{ad}})}, \qquad t_{\text{ad}} = \frac{E_{\text{ad}}}{2\pi k_{\text{B}}T}.$$

Low-T limit: $r_{\mathbb{Z}[\omega]}(1)/r_{\text{prism}}(1) = 6/2 = 3$; high-$T$ limit: $2(c/a)/\sqrt{3} \approx 1.88$. The Sazaki 2012 measurement of ≈ 2–3 near T_m is in the crossover.

4.6 QLL coefficient ξ

Lipowsky's quasi-liquid-layer thickness $d_{\text{QLL}}(T) = d_0 + \xi \log(T_m/\Delta T)$ is, upon substitution $u = \Delta T/T_m$, the leading UNDERWOOD log companion with base power $p = 0$. The higher-order UNDERWOOD coefficients $\xi_1, \xi_2, \ldots$ are new sub-leading corrections — see §5.

4.7 Proton activation energy E_A

$E_A = 0.575\,\text{eV}$ is the measured Arrhenius slope of $D(T)$ and is used as input, not derived. The *slope* match between Eq. (5) and experiment is therefore tautological; what is non-trivial is the prefactor D_0 (above) and its decomposition into $r(1) = 6$ and $N_{\text{ice}} = 8$.

4.8 Nakaya band count

The Nakaya diagram shows four habit bands between 0°C and −40°C: plates, needles, plates, columns. The framework reproduces the count via the doubling pattern $\Delta T_n = 3 \cdot 2^n - 2$ (see §6); no mock-modular form is required.

5 UNDERWOOD for the Log Branches

The QLL thickness and the Nakaya anisotropy both have logarithmic branches at $\Delta T \to 0$. UNDERWOOD, the author's augmented-basis framework from the ellipse-perimeter problem, is the rigorous tool for such functions.

Definition 5.1 (UNDERWOOD basis)**.** For a function $f : (0, 1] \to \mathbb{R}$ with a logarithmic branch at $\xi = 0$, the UNDERWOOD basis of order (n, K) is

$$\mathcal{B}_{n,K} \;=\; \{T_k(2\xi - 1) : 0 \le k \le n\} \;\cup\; \{\xi^{p+j} \log(1/\xi) : 0 \le j \le K\},$$

with Chebyshev polynomials T_k and base power p (the ellipse-perimeter paper uses $p = 2$; our QLL application uses $p = 0$).

Theorem 5.2 (Geometric convergence, Walsh–Bernstein separation)**.** *Under a log-branch regularity class [3], least-squares projection onto $\mathcal{B}_{n,K}$ satisfies $\|f - \hat{f}_{n,K}\|_\infty \leq C\,\rho_K^{-n}$ with ρ_K strictly increasing in K, and for any rational approximation $\hat{r}_P$ of matched parameter budget P, $\lim_{P\to\infty} \|f - \hat{r}_P\|_\infty / \|f - \hat{f}_{n,K}\|_\infty = \infty$.*

Application to QLL. Fitting Lipowsky-style $d_{\mathrm{QLL}}(T)$ data to $\mathcal{B}_{n,0}$ with $p = 0$ recovers ξ as the single log-companion coefficient. The in-plane-vs-vertical interpretation ambiguity that plagued earlier drafts of this paper dissolves: ξ is defined as d_0's log-branch coefficient, not as a geometric projection of a lattice rms displacement.

Application to Nakaya. UNDERWOOD approximates the full $r(\Delta T)$ curve near $\Delta T \to 0$ with provable geometric convergence. Mock modular forms are not required, and the sub-Kelvin predictions they would generate are replaced by UNDERWOOD's testable sub-leading coefficients $\xi_1, \xi_2, \ldots$ (§6).

6 The Nakaya Diagram

6.1 The doubling pattern

Nakaya's 1954 measurements and Libbrecht's subsequent refinements [6] identify three habit inversions at $\Delta T \in \{4, 10, 22\}$ K.

Observation 6.1 (Doubling)**.** The three observed band boundaries fit

$$\Delta T_n \;=\; 3 \cdot 2^n - 2, \qquad n = 1, 2, 3,$$

to 0.00% on all three data points simultaneously.

The sequence $(1, 4, 10, 22, 46, 94, \ldots)$ satisfies the linear recurrence $a_n = 2a_{n-1} + 2$, $a_0 = 1$, with no free parameters. Whether it derives from the Atkin–Lehner w_2 orbit on $\Gamma_0(2)$ remains an open question; we flag it as suggestive structure without committing to a mock-modular identification.

6.2 Predictions

Prediction 6.2 (Fourth habit band at -46°C)**.** $\Delta T_4 = 3 \cdot 2^4 - 2 = 46$ K gives $T = -46$°C. The predicted habit is lenticular or hourglass. Testable with controlled cold-chamber experiments.

Prediction 6.3 (D_2O dendrite peak shift)**.** The H-bond energies differ between isotopes: $E_{\mathrm{HB}}(H_2O) = 241$ meV, $E_{\mathrm{HB}}(D_2O) = 245$ meV (factor 1.0166). All Nakaya features scale linearly in E_{HB}; the D_2O dendrite peak is therefore predicted at

$$T_c(D_2O) \;=\; T_c(H_2O) \cdot 1.0166 \;=\; -10.10^\circ\mathrm{C},$$

a shift of $+4.30$°C. Testable by D_2O snow-crystal growth.

Prediction 6.4 (Sub-leading UNDERWOOD QLL coefficients)**.** An UNDERWOOD fit of interferometric $d_{\mathrm{QLL}}(T)$ with $K \geq 1$ yields non-zero $\xi_1, \xi_2, \ldots$ sub-leading Lipowsky corrections. Their sign and magnitude constrain microscopic QLL models.

7 Honest Tier Analysis

Eight observables of ice Ih, each labeled by proof tier:

Observable	Predicted	Measured	Tier
Proton E_A	0.575 eV (input)	0.575 eV	GENUINE (input)
Proton D_0	2.48×10^{-3} cm^2/s	2.5×10^{-3}	GENUINE (0.8%)
Bjerrum $E_L + E_D$ slope	1.02 eV (input)	1.02 eV	GENUINE (input)
Bjerrum prefactor	$36 = 6^2$	~ 36	GENUINE
Dendrite T_c	-16.25°C	-14.4°C	SEMI (1.85°C)
Dendrite ΔT_{dend}	2.99 K	~ 3 K	GENUINE (0.17%)
QLL ξ_0	UNDERWOOD-extracted	0.37 nm	GENUINE (post-UW)
Adatom basal/prism low-T	3	2–3	GENUINE
Nakaya band count	4	4	GENUINE
Nakaya band boundaries	$\{4, 10, 22\}$ K	$\{4, 10, 22\}$ K	EMPIRICAL PATTERN

What remains SEMI-GENUINE:

- T_c residual 1.85°C: the $R_{\text{res}} = 2\pi\sqrt{3}$ derivation is pure isotropic 2D; a c/a anisotropy correction would recover the missing degree. Open.
- Dendrite width scaling law $1/N_{\text{ice}}$ (chosen) vs $1/\sqrt{N_{\text{ice}}}$ (standard thermal broadening): the $1/N_{\text{ice}}$ choice gives the correct 3 K width but is not derived from first-principles stochastic dynamics. Open.

What remains EMPIRICAL: the doubling pattern (6.1). Observed to 0.00% on three data points; derivation via modular-group orbit suggestive but unproven.

8 Closure Plan

The framework is experimentally complete at the local level and open at two specific residuals. Four experimental and two theoretical closures would finish the programme.

C1 **UNDERWOOD fit of QLL data.** Interferometric $d_{\text{QLL}}(T_i)$ at ≥ 40 temperatures with $\lesssim 3$ Å noise. Produces rigorous ξ_0 with uncertainty and sub-leading ξ_1, ξ_2 if clean enough.

C2 **D_2O dendrite peak.** Measure the D_2O dendrite peak; compare to predicted -10.10°C. A shift in the $[+3, +6]$°C band from -14.4°C confirms the framework.

C3 **Fourth habit band test.** Grow ice at -46 ± 2°C under controlled supersaturation; document habit. Either confirms the doubling extrapolation or refutes it.

C4 **UNDERWOOD fit of Nakaya $r(\Delta T)$.** Digitize Libbrecht's full supersaturation sweeps. Fit to UNDERWOOD basis. Geometric convergence of residual validates the log-branch identification.

T1 **Derive the c/a anisotropy correction to R_{res}.** Expected to close the T_c 1.85°C residual by promoting the isotropic Eisenstein covolume $\sqrt{3}/2$ to the anisotropic 3D value with ice's specific $c/a = 1.6288$.

T2 **Derive the $1/N_{\text{ice}}$ width law.** A stochastic derivation in the spirit of Dicke superradiance or rearrangement-inequality cohomology. Expected to close the dendrite-width derivation.

Any two of {C1, C2, C3, C4} validate the framework experimentally; any one of {T1, T2} closes the remaining theoretical gaps.

9 What this Paper is and is Not

This paper IS. A one-document statement of everything the theta-mean framework predicts for ice Ih. Every formula traces to $E_{\mathrm{HB}} = 241\,\mathrm{meV}$, $r_{\mathbb{Z}[\omega]}(1) = 6$, and $N_{\mathrm{ice}} = 8$, plus the Onsager correlation factor 1/6 for proton diffusion and the UNDERWOOD basis for log-branch regression. Eight observables close or nearly close; residuals are labeled honestly.

This paper is NOT. A catalogue of speculative modular structure. An earlier preview proposed that the Nakaya morphology $r(\Delta T)$ is the real part of a specific weight-1 mock modular form on $\Gamma_0(6)$; that conjecture requires a q-expansion match which no one executed, and it is superseded here by UNDERWOOD's provably geometric approximation theory. The mock-modular identification is left as a suggestive hint at the doubling pattern's origin; nothing depends on it.

What the title promises. *How to Make Ice* is the recipe, fully stated by Eq. (1): three constants, the ice rule, the Eisenstein basal lattice, the UNDERWOOD toolkit for log branches. *Crystal Math* is §2. *Theta Means* is §3 (one theorem) plus bookkeeping throughout. These are the irreducible parts; what you see is what is there.

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